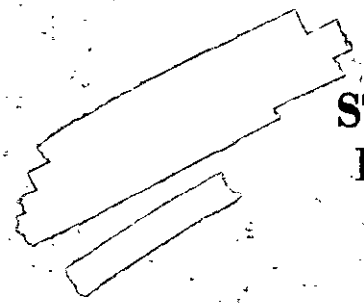
QQ–S–699
December 12, 1955

FEDERAL SPECIFICATION



STEEL SHEETS, CARBON, HOT-ROLLED, HEAVY GAGE STRUCTURAL QUALITY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers hot-rolled carbon-steel sheets 0.0449 inch and thicker in cut lengths or coils, intended for structural purposes where mechanical test values are required.

1.2 Classification.

1.2.1 *Classes.*—Hot-rolled carbon-steel sheets covered by this specification shall be of the following classes, as specified (see 6.2):

Class 1.—25,000 p. s. i. minimum yield point.
Class 2.—30,000 p. s. i. minimum yield point.
Class 3.—33,000 p. s. i. minimum yield point.

1.2.2 *Compositions.*—The steel sheets shall be of the following compositions, as specified (see 6.2):

Composition A.—Copper not specified.
Composition B.—Copper-bearing steel.

Unless otherwise specified in the contract or order, composition A shall be furnished.

1.2.3 *Finish.*—The steel sheets shall be hot-rolled unless hot-rolled-and-pickled is specified (see 6.2).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specification and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

QQ–M–151—Metals; General Specification for Inspection of.

Federal Standard:

FED–STD–No. 48—Tolerances for Steel and Iron Wrought Products.

Military Standard:

MIL–STD–163—Preparation of Steel Products for Domestic Shipment (Storage) and Overseas Shipment.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards through established distribution points in their agencies.

(Copies of Military standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The steel shall be made by either the open hearth or electric furnace process.

3.1.1 *Mill edges.*—Unless otherwise specified in the contract or order, sheets shall be furnished with mill edges (see 6.2).

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3.1.2 *Oiling.* — Unless otherwise specified in the contract or order, sheets shall be furnished free from oil or grease. Sheets ordered "oiled" shall have a thin film of mineral oil over the entire surface of each sheet (see 6.2).

3.2 Chemical requirements.—The chemical composition of the material shall conform to table I. A certified ladle analysis of each heat of steel shall be furnished by the contractor.

TABLE I.—*Chemical requirements*[1]

Classes 1, 2 and 3	Ladle analysis		Check analysis	
	Composition A	Composition B	Composition A	Composition B
	Percent	*Percent*	*Percent*	*Percent*
Carbon	0.25	0.25	0.31	0.31
Phosphorus	.04	.04	.050	.050
Sulfur	.05	.05	.063	.063
Copper	.33	.33	.35	.35
Copper (min.)	..	.20	..	.18

[1] Maximum unless otherwise indicated.

3.3 Tensile requirements. — The material shall conform to the tensile properties specified in table II.

TABLE II.—*Tensile requirements*

	Class 1	Class 2	Class 3
Tensile strength, min., p. s. i.	48,000	52,000	55,000
Yield point, min., p. s. i.	25,000	30,000	33,000
Elongation in 2 inch, min., percent: 0.2299 to 0.1450 inch, incl., in thickness	27.0	25.0	23.5
0.1449 to 0.0890 inch, incl., in thickness	24.5	23.0	21.5
0.0889 to 0.0610 inch, incl., in thickness	23.0	21.5	20.0
0.0609 to 0.0449 inch, incl., in thickness	21.0	20.0	18.5
Elongation in 8 inch, min., percent: 0.2299 to 0.1450 inch, incl., in thickness	20.0	18.5	17.5
0.1449 to 0.0890 inch, incl., in thickness	18.0	17.0	15.5

3.4 Bending.—The material shall not crack when tested in accordance with 4.4.2.3 to the following:

Class 1—Flat on itself.
Class 2—Around 1 thickness of the material.
Class 3—Around 1½ thicknesses of the material.

3.5 Dimensional and weight tolerances.—The material shall meet the ordered dimensions and weight and shall not vary from those ordered by amounts greater than those shown in the following paragraphs of Federal Standard FED. STD. No. 48:

Dimension and weight	Paragraph
Weight	11*a*1
Thickness	11*a*3
Width and length	11*a*4, 11*a*5, 11*a*6, and 11*a*8
Out of square	11*a*9
Resquared	11*a*10
Sheared circles	11*a*11
Camber	11*a*12
Flatness	11*a*13 and 11*a*14

3.6 Identification marking. — All material shall be identified with the specification number, class, composition and heat number legibly stenciled on the top sheet of each lift or shown on a tag attached to each coil or shipping unit. If sheets are shipped loose, the specification number, class, composition, and heat number shall be legibly stenciled on each sheet.

3.7 Workmanship.

3.7.1 *Sheets in cut lengths.* — The material shall be free from injurious laminations, blisters, slivers, open seams, pits from heavy rolled-in scale, ragged edges, holes, turned-down corners, or any other defects which may affect the serviceability of the material.

3.7.2 *Sheet in coils.*—If, upon use, the coils are found to contain unusable portions in excess of an amount specified in the contract or order (see 6.2), reimbursement shall be made for such unusable portion (see 4.3.3).

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—For purposes of chemical analysis and mechanical tests, a lot shall consist of all sheets of the same class and composition produced from the same heat of steel.

4.2 Sampling.

4.2.1 *Chemical analysis.*—One sample shall be taken by the Government inspector from each lot, and forwarded to a Government laboratory designated by the bureau or agency concerned (or to a laboratory acceptable to the Government inspector) for chemical analysis. If a lot is from a heat, part of which has been previously submitted in another lot and found to conform to 3.2, the Government inspector may waive the check analysis. The sample shall consist of at least 2 ounces of chips which are fine, and free from scale, oil, or other foreign matter.

4.2.2 *Mechanical tests.*—For the purpose of the tension and bend tests the Government inspector shall select sample sheets as specified in table III. One longitudinal tension test specimen and one transverse bend test specimen shall be taken from each sample sheet selected.

TABLE III.—*Sampling for mechanical tests*

Weight of lot	Number of samples[1]
Tons	
Up to 30, inclusive	1
31 to 50, inclusive	2
Over 50 to 100, inclusive	3
Over 100 to 150, inclusive	4

[1] If any one lot includes sizes which differ by 0.050 inch or more in thickness, one longitudinal tension test and one transverse bend test shall be made from the thickest and one longitudinal tension test and one transverse bend test from the thinnest material rolled. The minimum number of samples, for lots up to 30 tons, in this case, shall be 2.

4.3 Inspection.

4.3.1 *Dimensional and visual surface inspection.* — All material shall be subject to dimensional and visual surface inspection by the Government inspector, who shall satisfy himself as to whether the material meets the requirements of this specification. Lots containing defective material shall be subject to rejection.

4.3.2 *Reinspection.* — Lots rejected because of dimensional or visual surface defects may be resubmitted for inspection in accordance with 4.3.1, after the contractor has reworked and reinspected them to remove all nonconforming material.

4.3.3 *Sheets in coils.* — Since the physical shape of coils does not afford the same opportunity for surface inspection as is afforded for sheet in cut lengths, the presence of a minor number of imperfections in coils will not be cause for rejection (see 3.7).

4.4 Test procedures.

4.4.1 *Chemical analysis.*—The sample taken as specified in 4.2.1 shall be analyzed to determine conformance with table I.

4.4.2 *Mechanical tests.*

4.4.2.1 *Test specimens.* — Test specimens selected in accordance with 4.2.2 shall be prepared in accordance with Federal Specification QQ–M–151.

4.4.2.1.1 *Tension test specimens.* — Tension test specimens shall be taken longitudinally from the sheet as rolled and shall be of the full thickness of the material.

4.4.2.1.2 *Bend test specimens.* — The edges of the transverse bend test specimens shall be free from burrs. Filing or machining to remove burrs will be permitted.

4.4.2.2 *Tension test.*—The specimen shall be tested as specified in Federal Specification QQ–M–151, to determine compliance with table II. The yield point shall be determined by the drop of the beam, or by the divider method.

4.4.2.3 *Bend test.*—The transverse test specimens shall withstand being bent at room temperature with the crease of the bend parallel to the direction of rolling, through 180 degrees, as specified in 3.4 using a hand-operated vise with smooth jaws. Cracks developing at the edge of the specimen or coarse grain developing at the line of bend shall be disregarded.

4.5 Rejection.—If a specimen or sample fails to meet any of the requirements of this specifica-

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tion the lot represented by the specimen or sample shall be subject to rejection.

4.6 Retests.

4.6.1 If any test specimen or sample shows defective preparation or obvious lack of continuity of metal, the Government inspector may discard the specimen or sample and select another.

4.6.2 If the percentage of elongation of any tension test specimen is less than that specified in table II and any part of the fracture is outside the middle half of the gage length, as indicated by scribe marks on the specimen before testing, the test shall be discarded and a retest will be permitted.

4.6.3 If one test fails to meet the requirements specified and the purchaser permits a retest, two more tests shall be made from the same lot, in which case both tests shall conform to the requirements specified in this specification, otherwise the lot under test shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment not involving storage.—Material shall be prepared for shipment in accordance with commercial practice to insure carrier acceptance for safe delivery at destination at the lowest rate and shall meet, as a minimum, the requirements of carrier rules and regulations applicable to the mode of transportation.

5.2 Overseas shipment and domestic shipment involving extended storage. — Material shall be prepared for shipment in accordance with Military Standard MIL–STD–163.

5.3 Marking. — In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–163.

6. NOTES

6.1 Intended use. — Material furnished under this specification is intended for use in structural or load-bearing members. The level of the strength desired will govern the selection of the class to be used.

6.2 Ordering data.—Procurement documents should specify the following, and purchasers should exercise any desired options offered herein:

(a) Title, number, and date of this specification.

(b) Class and composition (see 1.2.1, and 1.2.2).

(c) Finish required (see 1.2.3).

(d) When sheared edges are to be furnished (see 3.1.1).

(e) When "oiled" sheets are to be furnished (see 3.1.2).

(f) When stretcher leveled standard of flatness is specified, whether grip or entry marks are to be inside or outside the specified length (see 3.5).

(g) When sheets in coils are ordered, the purchasing activity should specify the maximum permissible amount of unusable portions that will be allowed per coil (see 3.7.2).

(h) Size and quantity required.

(i) When "resquared" sheets are required.

(j) Whether sheets are to be prepared for domestic or overseas shipment and whether storage is involved (see section 5).

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—E O T
Navy—Sh A MC Or Y
Air Force.

U. S. GOVERNMENT PRINTING OFFICE: 1955

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PLATE NO. 15419 (BACK)

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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